Identification of Unknown B

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27 November 2023

CHEM 2081-003

Introduction

The goal of this lab is to identify an unknown compound by analyzing nuclear magnetic resonance (NMR), mass spectroscopy, and infrared spectroscopy data. These three methods are used in organic chemistry to distinguish the identity of specific compounds and analyze different unknown chemical compounds that have the same chemical formula. Identifying an unknown compound using these tools is necessary for the future if one is planning on pursuing a career in a laboratory.

Body

Infrared Spectroscopy (IR)

The wavelength from 2.5x10^-4 cm to 2.5x10^-3 cm is the area of analysis for IR. The position of signals is in units of wavenumber, which is the reciprocal of the wavelength and is expressed in cm^-1. IR is measured when energy is absorbed by the molecule, and specific bonds within the molecule become excited and start to vibrate. The amplitude of the vibration will increase or decrease based on if the frequency of the bond matches the frequency of radiation. A multitude of bonds within molecules will absorb different energies of infrared radiation which is why different molecules will not have the same absorption pattern.

There can be many different types of vibrations and bending within a molecule such as symmetrical or asymmetrical stretching as well as in-plane or out-of-plane bending. These different types of movements will lead to different energies of the molecule which ultimately leads to different absorption signals.

Percent transmittance is measured when infrared light is passed through a moving molecule, and is expressed as a function of frequency by an IR spectrometer. This data is then displayed in a chart that shows the frequencies absorbed by a vibrating molecule. Every group or structure on

the molecule has its own location and intensity on the absorption spectrum. Each signal is unique to certain bonds, structures, and groups within molecules so this information is ultimately used to determine the identity of the unknown compound.

Nuclear Magnetic Resonance Spectroscopy (NMR)

NMR helps with the understanding of the relationship between carbon atoms and hydrogen atoms in a compound by analyzing how the nuclei of atoms interact with each other and their electromagnetic radiation. NMR is the most useful tool when locating functional groups within a compound and displays multiple points of analysis of the compound. The signals on the spectrum are indicative of protons within a specific magnetic environment created by its surroundings. The location of this signal is important in compound analysis and is relative to the compound tetramethylsilane (TMS), which is used for reference of frequency of absorption. The location or "chemical shift" is the difference between the resonance frequency of the proton and that of the TMS divided by the operating frequency of the spectrometer. The location can also determine if a proton is shielded or unshielded based on whether it is to the right of the spectrum (upfield) or to the left of the spectrum (downfield). Shielded would be located upfield and unshielded would be located downfield.

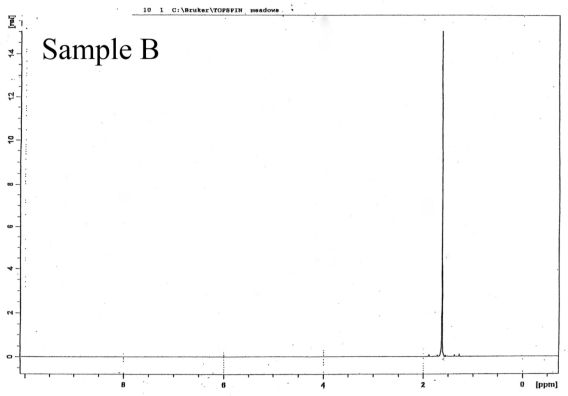
NMR is used with any nucleus that has a nuclear spin or odd mass number / atomic number. When the nucleus is placed on a strong magnetic field, it will align with the field because it acts as a positively charged magnet that spins. Due to the higher energy state, if a radio frequency is applied then the nucleus will align against the magnetic field. When the NMR is used, it emits a powerful bit of radio waves that flips the nuclear spin to the highest energy state. The decay of the signal is observed when it aligns back with the magnetic field and falls back into the lower energy state.

Mass Spectroscopy

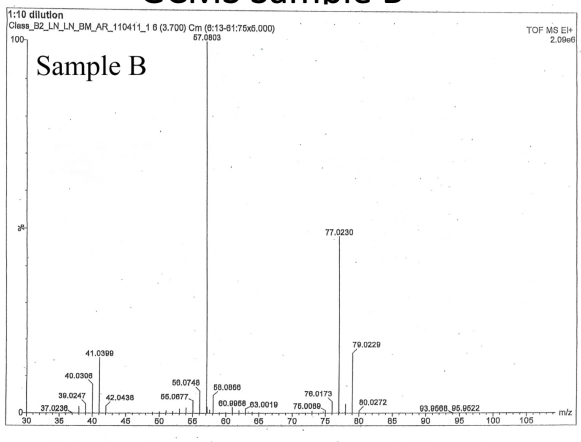
Mass spectroscopy is used to find the weight and molecular formula of a compound. It administers electron impact ionization, which is a process where the compound being assessed is hit with a high-energy electron and broken down into its ions and forms a radical cation intermediate known as a molecular ion. It is then sent to a magnetic field where deflected cations are detected and used to create a mass spectrum showing the relative abundance of each cation. When analyzing the mass spectrum, what indicates the molecular weight of the molecule is the peak of the molecular ion.

Data

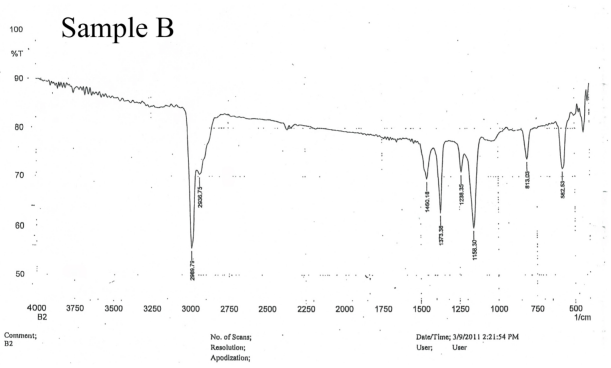
NMR Sample B



GCMS Sample B



FTIR Sample B



Analysis

Infrared Spectroscopy (IR)

Information about the functional groups can be collected upon looking at the IR. There are C-H alkane bonds at the peaks around 2990 and 2940. There are no significant peaks in the double / triple bond regions from 1600-1700. Lots of activity is occurring below 1500 cm^-1, which shows multiple single bonded carbon atoms. Between 1500-100 there are multiple methyl groups. Lastly, there can be a conclusion that there is a carbon to chlorine bond between the peaks at 813 and 582.

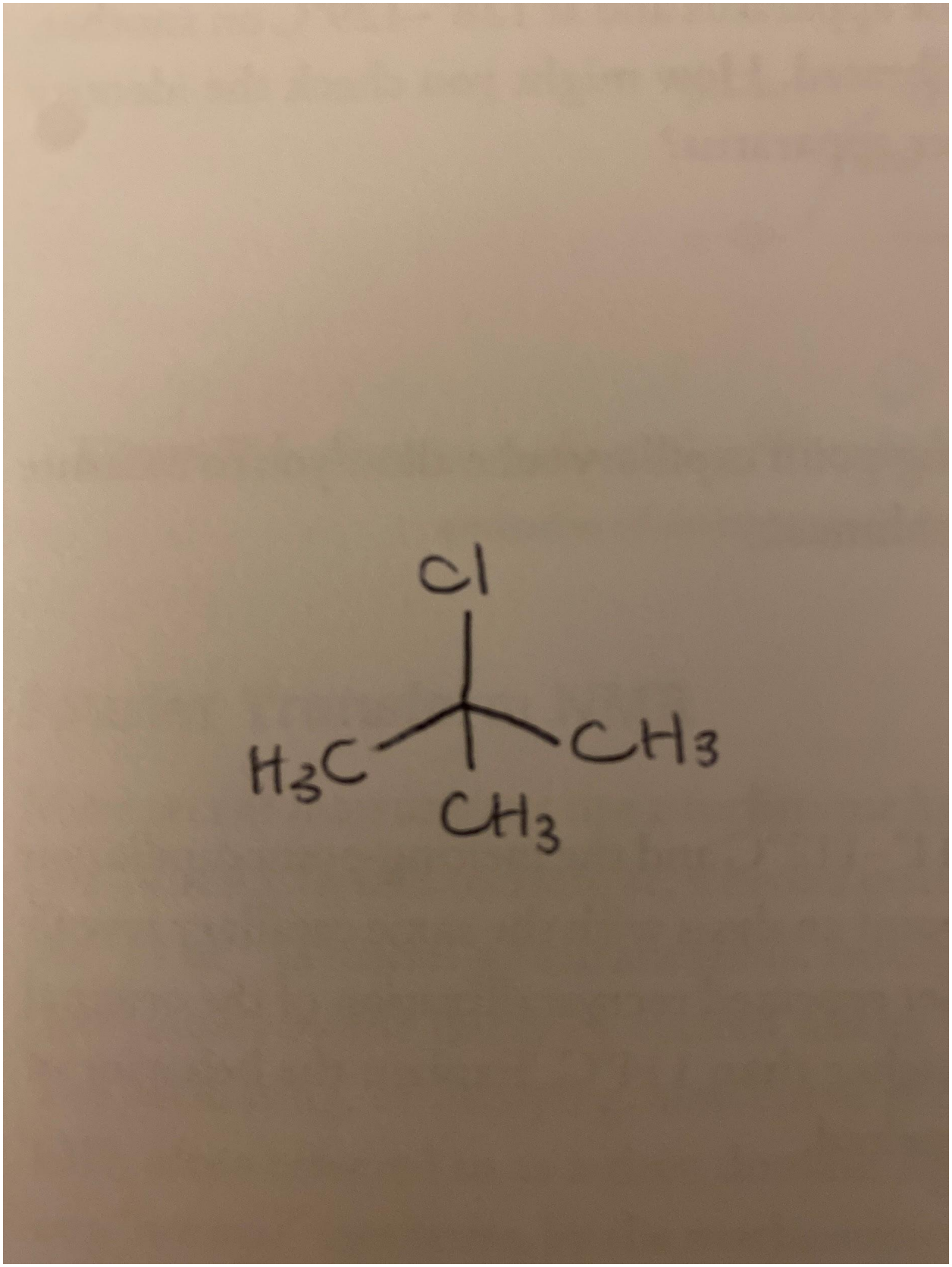
Nuclear Magnetic Resonance Spectroscopy (NMR)

There is a single peak between 1.5 and 2 ppm (around 1.6 ppm) for the unknown compound. Since there is only one peak, it can be inferred that there is only one type of proton in the compound. The peak that is located at 1.6 ppm indicates an electronegative atom attached to the compound. This results in a chemical shift downfield of the typical peak at about 1.2 ppm. This electronegative atom is most likely a halogen, such as chlorine. It is implied that there are multiple protons giving rise to the signal due to the fact that the peak is very tall with symmetrical distribution.

Mass Spectroscopy

The base peak, or largest peak, is located at 57.0803 m/z. The next two are located at 77 and 79 - the one at 79 is about ⅓ the height of the one at 77. The two things that can be concluded from these peaks: the most stable cation is C4H9 and a chlorine atom is present. The molecular ion peak can be matched with the molecular weight of t-butyl chloride since its peak is at 93.96.

Conclusion


Cl
H3C
CH3
CH3

The identification of unknown B has been found to be tert-butyl chloride. This conclusion has been reached by using the graphs from IR, NMR, and mass spectroscopy. All three graphs displayed the presence of chlorine attached to a carbon and that the compound is simple, consisting of only carbon, hydrogen, and chlorine with no special bonds or signals other than chlorine.

References

Keller L, Joseph K, and Miguel S. 2022 Fall. CHEM 2081 Organic Chemistry 1 Lab Manual [lab manual]. Auburn (Al): Auburn University.